50 Israeli inventions

STARTUP

You commented STARTUP, so here they are. Fifty things invented or built in Israel, each with what it does and who it helps. Every item is checked against the sources listed at the end.

Medicine and health

1. **PillCam (Given Imaging, Gavriel Iddan).** A camera in a swallowable capsule that films the small intestine. Helps patients skip invasive scopes and helps doctors find bleeding and Crohn's disease. [S1, S4]
2. **Copaxone (Weizmann Institute: Michael Sela, Ruth Arnon, Deborah Teitelbaum; Teva).** An immunomodulator for multiple sclerosis. Helps people with MS reduce relapses. [S1, S2]
3. **The Israeli emergency bandage (Bernard Bar-Natan, 1998).** A one-handed pressure bandage that stops severe bleeding. Helps soldiers, first responders and trauma patients. [S1, S2]
4. **ReWalk (Amit Goffer).** A motorized exoskeleton. Helps people with paraplegia stand and walk. [S1, S2]
5. **ExAblate (InSightec).** MRI-guided focused ultrasound that destroys tissue without cutting. Helps patients avoid open surgery. [S2]
6. **Azilect (Moussa Youdim and John Finberg, Technion; Teva).** A Parkinson's disease drug. Helps patients manage symptoms. [S1]
7. **CAR T foundations (Zelig Eshhar and Gideon Gross, Weizmann).** Chimeric antigen receptors, the basis of cell-based cancer immunotherapy. Helps blood cancer patients who ran out of options. [S1]
8. **Robotic spine surgery guidance (Mazor Robotics).** A robot that guides surgeons' tools in the spine. Helps surgeons work more precisely. [S1]
9. **WatchPAT (Itamar Medical).** A wrist and finger device for sleep apnea testing at home. Helps patients skip a night in a sleep lab. [S2]
10. **Deep TMS (Brainsway).** A helmet that applies deep magnetic brain stimulation. Helps people with hard-to-treat brain disorders. [S2]
11. **Elelyso, taliglucerase alfa.** An enzyme grown in carrot cells, FDA-approved in 2012. Helps people with Gaucher's disease. [S1]
12. **OrCam MyEye.** A small camera that reads text and recognizes objects aloud. Helps blind and visually impaired people. [S1]

Agriculture and food

13. **Modern drip irrigation (Simcha Blass and Yeshayahu Blass; Netafim, 1965).** Pipes with emitters that drip water straight to the roots. Helps farmers grow more with far less water. [S5, S6, S2]
14. **Long-life cherry tomatoes (Hebrew University: Nahum Keidar, Chaim Rabinovitch).** Tomato strains that stay fresh far longer. Helps growers and shippers cut waste. [S1, S2]
15. **AfiMilk (Kibbutz Afikim, 1977).** An electronic milk meter and herd management system. Helps dairy farmers track every cow. [S2]
16. **Hot-climate potatoes (Prof. David Levy, Hebrew University).** Potato strains that grow in heat and with salty water. Helps farmers in dry regions. [S2]
17. **Hybrid cucumber seeds (Esra Galun, Weizmann, 1950s).** Hybrid seed production without hand pollination. Helps seed companies and farmers raise yields. [S1]
18. **Grow Fish Anywhere (GFA).** Land-based fish farming in tanks with recycled water. Helps fish farmers far from the sea. [S1, S2]

19 **Bamba (Osem).** A peanut puff snack used in the research on early peanut introduction. Helps parents lower the risk of peanut allergy (ask your pediatrician). [S1]

Water

20 **Large-scale desalination (IDE Technologies).** Plants that turn seawater into drinking water. Helps dry countries secure their water supply. [S2]

21 **Mirtra trays (Tal-Ya Water Technologies).** Plastic trays that cover the soil around each plant's roots. Helps farmers cut watering and fertilizer, up to 50% by the maker's figures. [S1]

Tech and computing

22 **The USB flash drive, DiskOnKey (M-Systems, Dov Moran, 2000).** One of the first USB flash drives, sold with IBM. Helps everyone carry files in a pocket (credit is shared and disputed). [S7, S8, S1]

23 **ICQ (Mirabilis, 1996).** The first widely used instant messaging service. Helped millions chat online before social media. [S1, S2]

24 **The commercial stateful firewall (Check Point: Gil Shwed, Nir Zuk).** Software that inspects network traffic and blocks attacks. Helps companies protect their networks. [S1, S2]

25 **VoIP calling (VocalTec; Alon Cohen).** The first commercial internet phone calls. Helps people call worldwide for almost nothing. [S1]

26 **Intel 8088 (Intel Haifa).** The chip designed in Israel that powered the first IBM PC. Helped bring computers into homes and offices. [S1]

27 **EPROM (Dov Frohman, Intel, 1971).** A reprogrammable memory chip that keeps data without power. Helps engineers update devices. [S1]

28 **LZW compression (Abraham Lempel and Jacob Ziv, Technion, with Terry Welch).** A way to shrink files without losing data. Helps every computer store and send files faster. [S1]

29 **RSA encryption (Adi Shamir, with Ron Rivest and Leonard Adleman).** Public-key encryption. Helps keep online payments and messages secure. [S1]

30 **KVM virtualization (Avi Kivity, Qumranet, 2006).** Runs many virtual machines on one Linux server. Helps cloud providers run the internet's servers. [S1]

31 **3D depth sensing (PrimeSense).** Low-cost 3D vision for gesture control; Apple bought it in 2013. Helps phones recognize faces and devices see depth. [S1, S2]

32 **Digital printing press (Benny Landa, Indigo, 1993).** Prints straight from computer files. Helps printers do short, personalized runs. [S2]

33 **Objet 3D printers (now Stratasys).** Precise multi-material 3D printers. Helps designers and engineers prototype fast. [S2]

34 **Quicktionary (Wizcom Technologies).** A pen that scans and translates printed words. Helps language learners and travelers. [S1, S2]

35 **Powermat.** Wireless charging built into surfaces. Helps people charge without cables. [S2]

36 **Robomow (Friendly Robotics, 1998).** A robotic lawnmower. Helps homeowners skip the weekly mow. [S2]

Transport and mobility

37 **Waze (Ehud Shabtai, Amir Shinar, Uri Levine, 2008).** Crowd-sourced navigation with live traffic. Helps drivers save time; Google bought it in 2013. [S9, S2]

38 **Mobileye (founded 1999).** Camera-based driver assistance that warns of collisions. Helps drivers avoid crashes and carmakers build safer cars. [S2]

39 **Better Place (Shai Agassi).** An electric car charging and battery-swap network. A pioneer that showed what EV infrastructure could be (the company closed in 2013). [S2]

40 **Cormorant flying vehicle (Rafi Yoeli, Tactical Robotics).** A vertical take-off drone built to carry cargo and people. Helps rescue and supply in places roads can't reach. [S1, S2]

Security and defense

41 **Iron Dome (Rafael and IAI).** A mobile air defense system that intercepts rockets; first intercept in April 2011. Helps protect civilians under rocket fire. [S1]

42 **Trophy active protection.** Intercepts anti-tank rockets and missiles before they hit. Helps protect armored vehicle crews. [S1]

43 **Magshoe (IDO Security).** A step-on scanner that checks shoes at security. Helps travelers keep their shoes on. [S2]

44 **See-through-wall radar (Camro).** Ultra-wideband radar that senses people behind walls. Helps rescue teams and police. [S1]

45 **CornerShot (Lt. Col. Amos Golan).** A weapon accessory for seeing and aiming around corners. Helps police and soldiers stay out of the line of fire. [S1]

46 **Krav Maga (Imi Lichtenfeld).** A practical self-defense system. Helps people worldwide learn to protect themselves. [S1, S2]

Everyday life

47 **SodaStream.** Home carbonation machines. Helps families make sparkling drinks and skip plastic bottles. [S2]

48 **Epilady (1986).** The electric hair-removal device. Helps people skip waxing. [S2]

49 **Babysense (HiSense, 1992).** A monitor that alarms if a baby stops breathing. Helps parents sleep a little easier. [S2]

50 **Rummikub (Ephraim Hertzano).** The tile game for the whole family. Helps families put the phones down. [S1, S2]

Sources

S1: Wikipedia, "List of Israeli inventions and discoveries". https://en.wikipedia.org/wiki/List_of_Israeli_inventions_and_discoveries
S2: ISRAEL21c, "The top 64 innovations developed in Israel" (republished by Unpacked).
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S5: Wikipedia, "Simcha Blass". https://en.wikipedia.org/wiki/Simcha_Blass
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